

SEDAR 37: Southeastern U.S. Hogfish

Data Workshop

No specific recommendations were provided.

Assessment Workshop

Significant advancements in the understanding of life history for Hogfish were made since the last assessment in 2004 (SEDAR 6), mainly resulting from the effort of R. McBride and A. Collins (FWC-FWRI) and their collaborators from the fisheries, resulting in numerous publications and datasets. In particular, the age samples collected in both the WFL and FLK/EFL stock represent the vast majority of samples available for both stocks, providing for stronger estimates of growth and maturity than available from fisheries dependent sources or surveys.

While the life history is particularly well categorized in the WFL, where more research has focused (i.e., 2005-2007 life history study), questions still remain regarding the perceived differences in growth, maturity, and fecundity between the FLK/EFL and WFL stocks, and how these may be regulated by fishing pressures. In addition, life history studies and fisheries independent surveys are sorely needed for the GA-NC stock, particularly with respect to juveniles and mature females, since all available data is from fishery-dependent sources that catch primarily large, older males. Specific recommendations are as follows:

- (1) Conduct focused life history studies in the FLK/EFL and GA-NC stocks across a range of sizes/ages in order to test for differences in growth, maturity, and fecundity relative to the WFL stock where more information is available. While estimates from the FLK/EFL exist from the earlier life history study (1995-2001), additional sampling across a broader age spectrum by targeting more remote regions with lower fishing pressure (e.g., Dry Tortugas) may allow for better estimates of functional relationships.
- (2) Develop/improve fisheries-independent surveys for the GA-NC stock to specifically track Hogfish abundance. Currently, the SERFS video program only detects Hogfish in less than 5% of surveys, leading to difficulties in estimating abundance.
- (3) Improve biostatistical sampling of Hogfish in all regions from fisheries-dependent sources for both length and age observations.
- (4) Develop a life history study to ascertain the contribution of males to spawning reproductive potential (SRP). Appropriate determination of male contribution will provide more certainty in modeling reproduction, which has a strong influence on stock status and could be instrumental in designing appropriate management regulations with respect to size limits to protect the spawning biomass.

Review Process

Research Recommendations from the CIE Reviewer Reports

R.I.C. Chris Francis Recommendations

Consider the research recommendations provided and make any additional recommendations or prioritizations warranted.

a) Clearly denote research and monitoring that could improve the reliability of, and information provided by, future assessments.

The Assessment Report presented four research recommendations, which can be summarised as follows.

1. More growth, maturity, and fecundity information for FLK/EFL and GA-NC
2. Fishery-independent surveys for GA-NC
3. Improve age and length sampling of hogfish fisheries
4. Contribution of males to spawning reproductive potential

Of these, I think 3 and 2 are most likely to improve future assessments. With regard to 3, I would strongly emphasize the age data (age compositions are much more informative in a stock assessment model than length compositions) and comment that doubling the number of trips sampled has a much greater effect on precision than doubling the number of fish sampled per trip. There is a need to check that the ageing of the sampled fish is consistent with the validated method (see Section 3.1a) and to develop reliable methods to construct length compositions from these data (see Section 3.1a). With regard to 2, I note that GA-NC was the weakest of the three assessments because there was only one biomass index, and this was not well fitted by the model (see AR Figure 11.2.1.2.16).

I think that 4 is important, but perhaps more to inform fishery management (e.g., are special measures needed to protect males?) than to improve the stock assessment.

A comparative study of alternative methods of biomass index calculation would be useful. I was startled by the array of different methods used for indices in these assessments (markedly different methods are described in Background Documents 37-02, 37-05, 37-09, and 37-12) and wondered whether it was either necessary or desirable to use so many different methods. Perhaps some rationalization of methods would be possible. The study should include an investigation of sensitivity to habitat classification (see Section 3.1b) and the desirability of including year interactions (see beginning of Section 3.1) and, for the survey data, design-based methods (because they require fewer assumptions I think these methods are preferable unless demonstrated to be markedly less precise than the model-based methods).

Background Document 37-01 showed that fish from east Florida waters were genetically distinct from those from North Carolina waters, but was unable to be specific about the location of the boundary between these stocks. Samples from Georgia and South Carolina would be useful to check the stock assessment assumption that this boundary lies at the Florida-Georgia border.

b) Provide recommendations on possible ways to improve the SEDAR process.

I have no recommendations to make but would like to comment that I think the present review would have been much better informed, and thus more useful, had it involved participation in an assessment review meeting, during which both discussions with the assessment team and some additional model runs could have resolved some important uncertainties for me.

Improvements for next assessment

Provide guidance on key improvements in data or modeling approaches which should be considered when scheduling the next assessment.

Three sets of data merit particular attention in the next assessment. For length compositions, improve their construction (Section 3.1a), and selections (i.e., dropping years with small [unrepresentative] sample sizes – see Section 3.1d), and improve their weighting (Section 3.1c). Initial equilibrium catches need to be better related to pre-1986 catches (Section 3.1a). For recreational catches, use small SEs, smooth spikes, and construct alternative low and high catch histories (Section 3.1c) to use in sensitivity analyses.

Two other model assumptions are worth reconsidering: the default selectivity (Section 3.0.2), and the growth parameters (including CVs) used for each stock (Section 3.0.3).

Some other suggestions that might improve the next assessment are as follows.

- Use a likelihood profile on R_0 to help understand the data sets affecting the estimation of this parameter, and thus the robustness of the assessments' estimates of initial depletion (Section 3.0.1).
- Use sensitivity analyses to investigate the effect of uncertainty in steepness, just as was done for natural mortality (Section 3.5a).
- Characterize uncertainty in the projections by repeating them for selected sensitivity analyses (Section 3.5a).

Recommendations

Amongst the Assessment Report's four suggestions for future research I recommend giving priority to the two concerning improve age and length sampling of hogfish fisheries (the focus should be on ages) and fishery-independent surveys for GA-NC. In addition, I recommend (a) research to rationalize the calculation of biomass indices, and (b) genetic sampling to better identify the location of the stock boundary between east Florida and North Carolina. (See Section 3.6a for more details).

For the next hogfish assessment I recommend that particular attention be paid to three data sets: length compositions (construction, selection, and weighting), initial equilibrium catches; and the treatment of uncertainty in recreational catches. Two hypotheses that should be reconsidered are those concerning default selectivities and growth. Other recommendations include a likelihood profile on R_0 , a sensitivity analysis for steepness, and a way of characterizing uncertainty in the projection results. (See Section 3.7 for more details).

Medley Recommendations:

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a.) Clearly denote research and monitoring that could improve the reliability of, and information provided by, future assessments.

The assessment document provides useful recommendations for further research, covering research on hogfish biology and improved monitoring data. Research on life history and growth has provided a good basis for the stock assessment modelling. While on-going research on hogfish biology will be useful, it is not a critical area for reducing uncertainty in the assessment at this stage. Improvements in monitoring data are more important.

Good stock assessments will not be possible without good estimates of catches and abundance indices. The assessment recommendations consist of improvements in biological sampling for

lengths and age across all fisheries, and development of a fishery independent abundance index for the GA-NC stock.

While the assessment report recommendations are important, other areas of the assessment will also require improvement. The most valuable improvement would likely be better recreational catch data reporting. The proportional standard errors are very high for all estimated landings and it seems unlikely that catches will vary so significantly year by year as currently estimated. Some of these problems are historical, and recent years' catches appear more accurate. Dealing with past errors is an issue of improved robust estimation only, whereas ongoing improved sampling and estimation procedures could reduce errors in future. With recreational catches being so high in many Florida fisheries, improvement in monitoring recreational catches should provide benefits to a wide number of fishery assessments.

b) Provide recommendations on possible ways to improve the SEDAR process.

The SEDAR process would benefit from greater guidance on assessment output and greater focus on assessing uncertainty. It is recommended that the SEDAR process include:

- The stock assessment should identify a pair of sensitivities to bracket the uncertainty and bootstrap or MCMC simulations should be applied to these as well as the base case. These uncertainties should be included in the projections.
- Sensitivities should report changes in stock and fishing status, not only changes in parameter estimates. Parameter estimates may be correlated, so important indicators (e.g. F_{2012}/F_{MSY} , SSB_{2012}/SSB_{MSY}) may change very little.
- The assessment should report the breakdown of negative likelihood contributions for each of the main data components.
- It is useful to provide the input data and results in spreadsheet or text form if possible, so that additional graphs and tables can be made if necessary as part of the review. Although in most, but not all cases, tables are provided in the report and data can be extracted from these with some effort, it would be easier if original information was provided. Further diagnostic plots, such as observed vs expected values, residual plots and so on would have been useful and some of the presented graphs were unclear. Information provided in text or spreadsheets allows reviewers to examine what they want while avoiding unnecessary work for the assessment team.

Terms of reference for the stock assessment and this review might be improved and better aligned. Specifically, the stock assessment ToRs should require that uncertainty is included in the projections, which is implied in the Review ToR 4.

The assessment should be given more guidance on practical management interventions so that the projection can be based on real options. In this case, it is also unclear how the fishing mortality targets used in the projections might be implemented where catches are so poorly monitored.

Provide guidance on key improvements in data or modeling approaches which should be considered when scheduling the next assessment.

It may be better to fit the model to the total recreational fishing effort (angler days) rather than

the catch directly (e.g. Porch et al. 2006). The year-to-year variation in effort is much lower than the estimated catches and probably provides a better estimate of the variation in fishing mortality. An explicit likelihood linking the intercept samples and the total catch can be included in the stock assessment. This would account for the sampling error explicitly, and allow the model to smooth through the catches providing more accurate estimates.

Currently, with no other information, the catch in the model is likely to follow the input estimate, while the implied catchability is not consistent with the abundance indices. If estimated within the model, the catches would be smoothed, but probably more accurate.

Including the catch estimation within the stock assessment is desirable, but may be too onerous as it would probably require developing a bespoke model. An alternative might be to link catch estimation to the development of abundance indices, which would limit the year to year variation in catch rates. Linear models could be used to build catch estimates conditional on observations across years consistent with the abundance indices implied catch rates rather than as independent samples.

Whereas parametric bootstraps provide an excellent tool for estimating uncertainty, the method used here does not account for much of the known uncertainty in input values. It would probably be better to simulate bootstrap datasets externally to SS3 where uncertainty in the dataset could be more accurately modelled. For example, the MRIP/MRFSS derived catch estimates could themselves be bootstrapped to generate alternative catch time series.

Identifying ways to remove the retrospective bias for the GA-NC and WFL stock assessments should help identify primary sources of structural error. Estimating time varying catchability is difficult within the model, but external adjustments to input data based on likely changes in catchability as well as adjusting catches (e.g. applying a smoothed catch time series) could at least identify possible causes for the bias as well as provide alternative sensitivities.

The purpose of sensitivity runs should not be so much to determine possible ranges for parameter estimates, but to try to incorporate uncertainty in key assumptions into management advice. The aim should be to identify a reasonable range from the sensitivities to capture this uncertainty and include the additional model configurations in projections.

Future additional sensitivities should be considered and include:

- Apply more changes on data component weights (lambdas) to explore how they affect the assessment outcome. Specifically for the GA-NC stock, weights to force fits alternately to the abundance index, landings and length frequency data (use the “continuous Fs” option) should help elucidate problems in this model.
- Time varying selectivity could improve abundance indices, including the WFL and FLK/EFL commercial spear, and the GA-NC commercial hook and line. The RVC Keys index selectivity should be split into two separate series before and after 2000, or possibly drop the earlier period from the assessment.
- Nominal indices and indices based only on non- $^{\circ}\text{C}$ -zero trips were not tried. It is not clear from the information presented how much influence the APC / binomial model has on the final index. It may be useful to consider the positives trips model alone (hogfish caught ≥ 1) as this could

avoid bias in the trip selection procedure which is always very uncertain. If these alternative abundance indices give different

Tingley Recommendations:

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b) Provide recommendations on possible ways to improve the SEDAR process.

a) Research and monitoring recommendations:

- In a fishery with multiple data deficiencies, one of the potential objects of modeling is to identify those datasets that, by their inadequacy, associated uncertainties or absence, have a disproportionate impact on the outcomes of the assessment that managers have a particular interest in. This can then provide a coherent input to the prioritization of future research effort aimed at improving the assessment most effectively. More effective progress may be made by using the model outputs to review the immediate future research focus and prioritization.

Four recommendations are made in the assessment document but are not prioritized. The prioritization recommended by the reviewer of these, is as follows:

- 1) Improve the biostatistical sampling of hogfish.
- 2) Develop/ improve the fishery-independent surveys for the GA-NC stock and fisheries.
- 3) Conduct focused life history studies for the FLK/EFL and GA-NC stocks.
- 4) Develop a life-history study to address male contribution to spawning reproductive potential.

The first two of these recommendations are by far the more important, as these sit right at the heart of delivering acceptable stock assessments for these stocks.

b) SEDAR process improvement recommendations:

The organizational approach, provision of clear ToR and provision of documents for the SEDAR process is of a very high standard. The recommendation that follows addresses minor issues that particularly address the needs of external reviewers and general readers alike in understanding these fisheries and the complex assessments in a relatively short space of time.

It is recommended that the following issues be considered for inclusion in future SEDAR assessment reports.

- A report structure with fully consecutive page numbering would have made the reading and reviewing the report easier.
- A list of acronyms should be included in the report. There was no list of acronyms in the assessment report which, given the number of acronyms used, would have been very useful and would have expedited the work of the review (see for example

http://sero.nmfs.noaa.gov/sustainable_fisheries/more_info/documents/pdfs/glossary_of_fishery_terms.pdf).

- Tables in assessment reports need to be appropriately formatted to enable effective interpretation of their information content: in a number of key tables in the assessment

report, the columns of figures were neither right-justified nor aligned at the decimal point and the numbers had variable decimal places. The numbers in some of the tables were overly precise (i.e. there are too many places of decimals). While minor in themselves, these make reading the tables for scale, errors, outliers and areas of transition, both slower and harder.

- A map describing the key stock areas, locations and boundaries referred to, would have aided the reader's understanding of the spatial context of the fisheries, the stock structure, and sampling locations referred to.

Provide guidance on key improvements in data or modeling approaches which should be considered when scheduling the next assessment.

- It is recommended that there should be a specific requirement for assessment teams to consider and report on the quality of all time-series being considered as abundance indices. This consideration should take account of evidence, or if evidence is lacking, logical arguments that support or oppose the likelihood of each time-series indexing the stock in question. Where time-series are found to be unlikely or highly unlikely to index abundance they should be omitted from the assessment. Where evidence and/or logical arguments are inconclusive, the value of the time-series may be addressed through running sensitivities. Specific objectives of this approach should be to (i) raise the quality standard of the input data to help improve the fit of the model to the data and increase the robustness of the assessment; and (ii) to eliminate the inclusion of times-series that show opposing trends in abundance within the same model run where the time-series relate to the whole stock or the same stock components.
- Length frequency and age samples from these fisheries are of borderline quality for enabling adequate stock assessments for the three hogfish stocks to be developed.

Sampling in more recent years has been better than that from earlier years but ideally should be improved further. It is recommended that a more consistent approach to obtaining sufficient samples that are representative of each fishery in each year be developed. An approach that aims to obtain a balance of samples from the different fisheries and stocks should be developed and implemented. This will help enable stock assessments of adequate quality to be developed in future. This is essentially the same as the recommendation to 'improve the biostatistical sampling of hogfish' made by the assessment team (see section 6 above).